

Assignment No. 4

Azure Cloud Fundamentals and Data Pipeline Implementation using Azure Data Factory

Celebal Technologies (CEI Internship)

Name: Ritesh Deshmukh

Assignment No.: 4

Title: Azure Cloud Fundamentals and Data Pipeline Implementation using Azure Data Factory (ADF)

Objective

To understand Microsoft Azure cloud concepts and build an end-to-end data pipeline using Azure Blob Storage and Azure Data Factory. The assignment demonstrates how to create Azure resources, upload datasets to Blob Storage, configure Azure Data Factory, validate file metadata, copy data between Blob containers, and monitor pipeline execution.

Tools Used

- Microsoft Azure Portal
 - Azure Resource Group
 - Azure Storage Account
 - Azure Blob Storage
 - Azure Data Factory (ADF)
 - Azure IAM (Role-Based Access Control)
-

Dataset

Dataset Name: Sample - Superstore.csv

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer	Customer Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
1	CA-2016-1	6/9/2014	6/9/2014	Standard	Clare Gut	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-1F	Furniture	Bookcases	Bush Sonnet	261.96	2	0	41.9136
2	CA-2016-1	6/9/2014	6/9/2014	Standard	Clare Gut	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-1F	Furniture	Chairs	Hon Delux	731.94	3	0	219.582
3	CA-2016-1	6/9/2014	6/9/2014	Standard	Darin Van	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-1O	Office Sup	Labels	Self-Adhes	14.62	2	0	6.8714
4	US-2015-1	6/9/2014	6/9/2014	Standard	Sean O'Do	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-1C	Furniture	Tables	Bretford C	957.5775	5	0.45	-383.031
5	US-2015-1	6/9/2014	6/9/2014	Standard	Sean O'Do	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-1O	Office Sup	Storage	Eldon Fold	22.368	2	0.2	2.5164
6	CA-2014-1	6/9/2014	6/9/2014	Standard	Brosina Hc	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-1C	Furniture	Furnishings	Eldon Expr	48.86	7	0	14.1694
7	CA-2014-1	6/9/2014	6/9/2014	Standard	Brosina Hc	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-1C	Office Sup	Art	Newell 32i	7.28	4	0	1.9656
8	CA-2014-1	6/9/2014	6/9/2014	Standard	Brosina Hc	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-1C	Technology	Phones	Mitel 532C	907.152	6	0.2	90.7152
9	CA-2014-1	6/9/2014	6/9/2014	Standard	Brosina Hc	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-1O	Office Sup	Binders	DRI Angle	18.504	3	0.2	5.7825
10	CA-2014-1	6/9/2014	6/9/2014	Standard	Brosina Hc	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-1O	Office Sup	Appliances	Belkin F5C	114.9	5	0	34.47
11	CA-2014-1	6/9/2014	6/9/2014	Standard	Brosina Hc	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-1C	Furniture	Tables	Chromcraft	1706.184	9	0.2	85.3092
12	CA-2014-1	6/9/2014	6/9/2014	Standard	Brosina Hc	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-1C	Technology	Phones	Konitel Z9	911.424	4	0.2	68.3568
13	CA-2017-1	6/9/2014	6/9/2014	Standard	Andrew Al	Consumer	United States	Concord	North Carolina	28027	South	OFF-PA-1O	Office Sup	Paper	Xerox 156	15.552	3	0.2	5.4432
14	CA-2016-1	6/9/2014	6/9/2014	Standard	Irene Mad	Consumer	United States	Seattle	Washington	98103	West	OFF-BI-1O	Office Sup	Binders	Fellowes P	407.976	3	0.2	132.5922
15	US-2015-1	6/9/2014	6/9/2014	Standard	Harold Pax	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-AP-1O	Office Sup	Appliances	Holmes Re	68.81	5	0.8	-123.858
16	US-2015-1	6/9/2014	6/9/2014	Standard	Harold Pax	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-BI-1O	Office Sup	Binders	Stores Dur	2.544	3	0.8	-3.816
17	CA-2014-1	6/9/2014	6/9/2014	Standard	Pete Kriz	Consumer	United States	Madison	Wisconsin	53711	Central	OFF-ST-1O	Office Sup	Storage	Stur-D-Sto	665.88	6	0	13.3176
18	CA-2014-1	6/9/2014	6/9/2014	Standard	Alejandro C	Consumer	United States	San Jose	Utah	84084	West	OFF-ST-1O	Office Sup	Storage	Fellowes S	55.5	2	0	9.99
19	CA-2014-1	6/9/2014	6/9/2014	Standard	Zuschuss E	Consumer	United States	San Francisco	California	94109	West	OFF-AR-1C	Office Sup	Art	Newell 34i	8.56	2	0	2.4824
20	CA-2014-1	6/9/2014	6/9/2014	Standard	Zuschuss E	Consumer	United States	San Francisco	California	94109	West	TEC-PH-1C	Technology	Phones	Cisco SPA	213.48	3	0.2	16.011
21	CA-2014-1	6/9/2014	6/9/2014	Standard	Zuschuss E	Consumer	United States	San Francisco	California	94109	West	OFF-BI-1O	Office Sup	Binders	Wilson Jor	22.72	4	0.2	7.384
22	CA-2016-1	6/9/2014	6/9/2014	Standard	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AR-1C	Office Sup	Art	Newell 33i	19.46	7	0	5.0596
23	CA-2016-1	6/9/2014	6/9/2014	Standard	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AP-1O	Office Sup	Appliances	Acco Six-G	60.34	7	0	15.6884
24	US-2017-1	6/9/2014	6/9/2014	Standard	Sandra Fla	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-CH-1F	Furniture	Chairs	Global Del	71.372	2	0.3	-1.0196
25	CA-2015-1	6/9/2014	6/9/2014	Standard	Emily Burn	Consumer	United States	Orem	Utah	84057	West	FUR-TA-1C	Furniture	Tables	Bretford C	1044.63	3	0	240.2649

The dataset was uploaded into the **input** Blob container and used as the source file for building the Azure Data Factory pipeline.

Task 1: Resource Group Creation

A Resource Group was created to organize all Azure resources required for this assignment.

Screenshot:

portal.azure.com/?Microsoft_Azure_Education_correlationId=dc50ed97-fa76-479b-869f-99b34ff1112e&Microsoft_Azure_Educat...

Microsoft Azure

Search resources, services, and docs (G+)

Home > Resource Manager

Resource Manager | Resource groups

Default Directory

How to manage changes with deployment tools? Export resource groups using Bicep or Terraform +1

Search

+ Create Manage view Refresh Export to CSV Open query Assign tags Group by none

Filter for any field... Subscription equals all Location equals all Add filter

No resource groups to display

Resource groups provide a logical container to manage and organize Azure resources, simplifying administration and enabling efficient resource management.

+ Create

Learn more

Showing 1 - 0 of 0. Display count: auto

Give feedback

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Home > Resource Manager | Resource groups

Create a resource group

Basics Tags **Review + create**

[Automation Link](#)

Basics

Subscription	Azure for Students
Resource group name	CelebalWeek4
Region	Central India

Tags

None

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Task 2: Storage Setup

An Azure Storage Account was created successfully.

Two Blob containers were created:

- **input** – Stores the source dataset.
- **destination** – Stores the copied output dataset.

The **Sample - Superstore.csv** file was uploaded to the **input** container.

Screenshot:

Microsoft Azure

Search resources, services, and docs (G+)

Home > Storage center | Blob Storage

Create a storage account

Basics Advanced Networking Data protection Security Encryption Tags **Review + create**

[View automation template](#)

Basics

Subscription	Azure for Students
Resource group	CelebalWeek4
Location	Central India
Storage account name	riteshstorage001
Primary service	Azure Blob Storage or Azure Data Lake Storage
Performance	Standard
Replication	Locally redundant storage (LRS)

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Microsoft Azure

Search resources, services, and docs (G+)

riteshdeshmukh0015@...
DEFAULT DIRECTORY (RITESHDES...

Home

riteshstorage001_1783875968672 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : riteshstorage001_178387... Start time : 7/12/2026, 10:36:25 PM
Subscription : Azure for Students Correlation ID : aa5a54de-b12f-4a22-afa...
Resource group : CelebalWeek4

> Deployment details

> Next steps

Go to resource

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.
Set up cost alerts >

Microsoft Defender for Cloud

Secure your apps and infrastructure
Go to Microsoft Defender for Cloud >

Home > riteshstorage001

riteshstorage001 | Containers

Storage account

Search

+ Add container Upload Refresh Delete Change access level Restore containers Edit columns

Search containers by prefix

Only show active containers

Showing all 2 items

☐	Name	Last modified	Anonymous access level	Lease state	Storage connector
☒	\$logs	12/07/2026, 22:36:50	Private	Available	-
☐	destination	12/07/2026, 22:39:02	Private	Available	-

Microsoft Azure

Search resources, services, and docs (G+)

riteshdeshmukh0015@...
DEFAULT DIRECTORY (RITESHDES...

Home > riteshstorage001 | Containers

input

Container

Search

+ Add Directory Upload Change access level Refresh Delete Copy Paste Rename

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

input

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

Successfully uploaded blob(s)
Successfully uploaded 1 blob(s).

Task 3: Azure Data Factory Setup

An Azure Data Factory instance was created successfully.

The following components were configured:

- Linked Service (Azure Blob Storage)
- Source Dataset
- Destination Dataset
- Get Metadata Activity

Screenshot:

The image displays two screenshots of the Azure portal's 'Create Data Factory' wizard.

Top Screenshot: Project details

Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ Azure for Students
Resource group * ⓘ CelebalWeek4
[Create new](#)

Instance details

Name * ⓘ adf-ritesh001 ✓
Region * ⓘ (Asia Pacific) Central India
Version * ⓘ V2

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Bottom Screenshot: Basics

Basics

Subscription	Azure for Students
Resource group	CelebalWeek4
Name	adf-ritesh001
Region	Central India
Version	V2

Networking

Connect via	Public endpoint
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riteshdeshmukh0015@...
DEFAULT DIRECTORY (RITESHDES...

Home

Microsoft.DataFactory-20260712225624 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Deployment succeeded

Deployment 'Microsoft.DataFactory-20260712225624' to resource group 'CelebalWeek4' was successful.

Pin to dashbo... Go to resource gr...

✓ Your deployment is complete

Deployment name : Microsoft.DataFactory-... Start time : 7/12/2026, 10:58:21 PM
Subscription : Azure for Students Correlation ID : 843bbcab-7c06-4de5-b...
Resource group : CelebalWeek4

Deployment details

Next steps

Go to resource

Give feedback

Tell us about your experience with deployment

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riteshdeshmukh0015@...
DEFAULT DIRECTORY (RITESHDES...

Home > Microsoft.DataFactory-20260712225624 | Overview

CelebalWeek4

Resource group

Generate Bicep code to duplicate this resource group. Help me generate Terraform for this resource group configuration. +1

Search

+ Create Manage view Delete resource group Refresh Export to CSV Open query ... Group by none

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Cost Management

Monitoring

Automation

Essentials

JSON View

Resources Recommendations

Filter for any field... Type equals all Location equals all Add filter

Name ↑	Type	Location
adf-ritesh001	Data factory (V2)	Central India
riteshstorage001	Storage account	Central India

Add or remove favorites by pressing Ctrl+Shift+F

Microsoft Azure

Data Factory adf-ritesh001

Search

riteshdeshmukh0015@gmail.com
DEFAULT DIRECTORY

Plan ahead with Fabric

Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. Start exploring

Azure Data Factory allows you to configure a Git repository with either Azure DevOps or GitHub. Git is a version control system that allows for easier change tracking and collaboration.

Learn more

Set up code repository

Author

Data factory

adf-ritesh001

New

Ingest

Copy data at scale once or on a schedule.

Orchestrate

Code-free data pipelines.

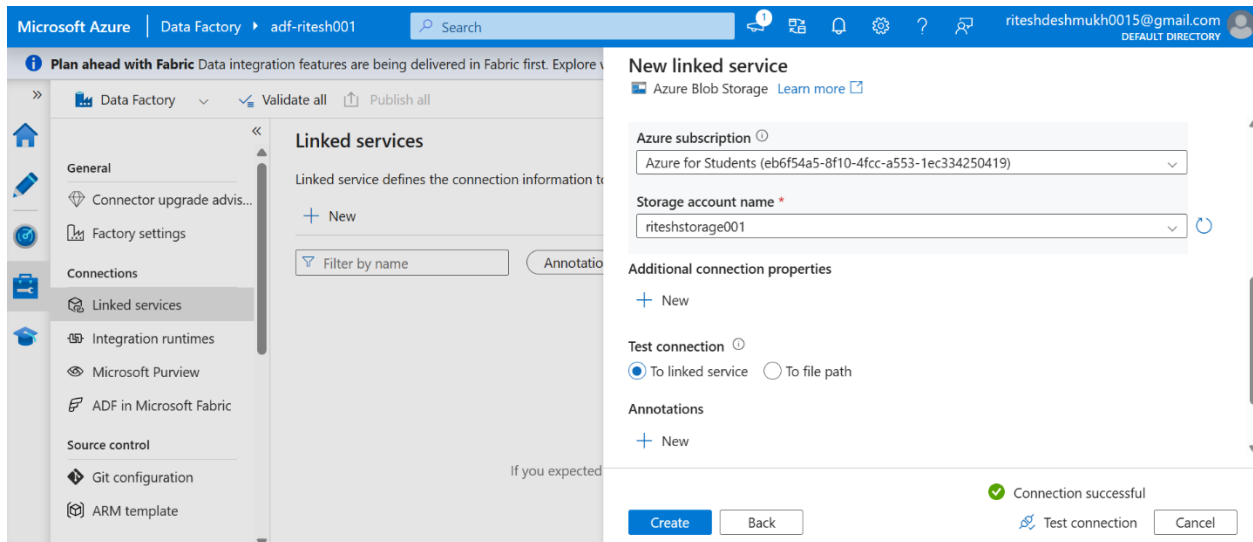
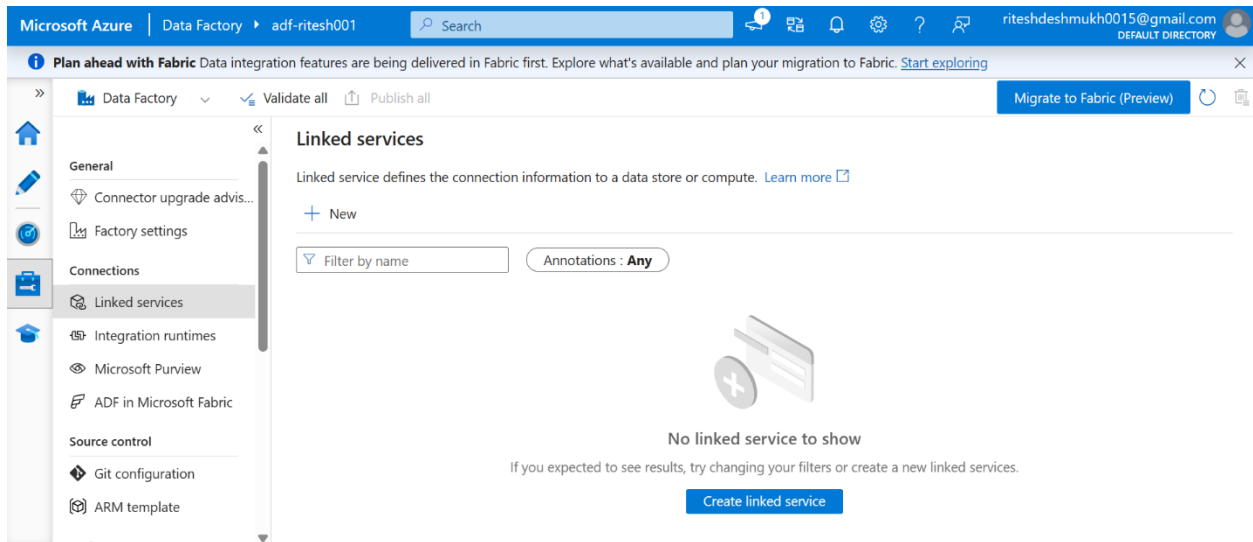
Transform data

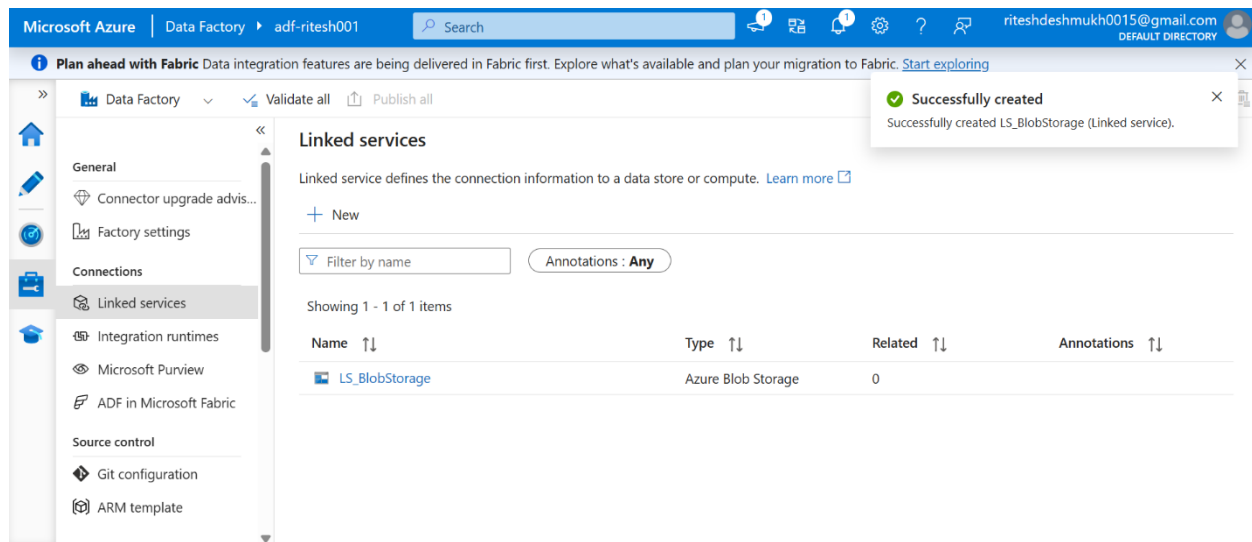
Transform your data using data flows.

Configure SSIS

Linked Service

A Linked Service was created to establish a connection between Azure Data Factory and Azure Blob Storage using the Storage Account credentials.





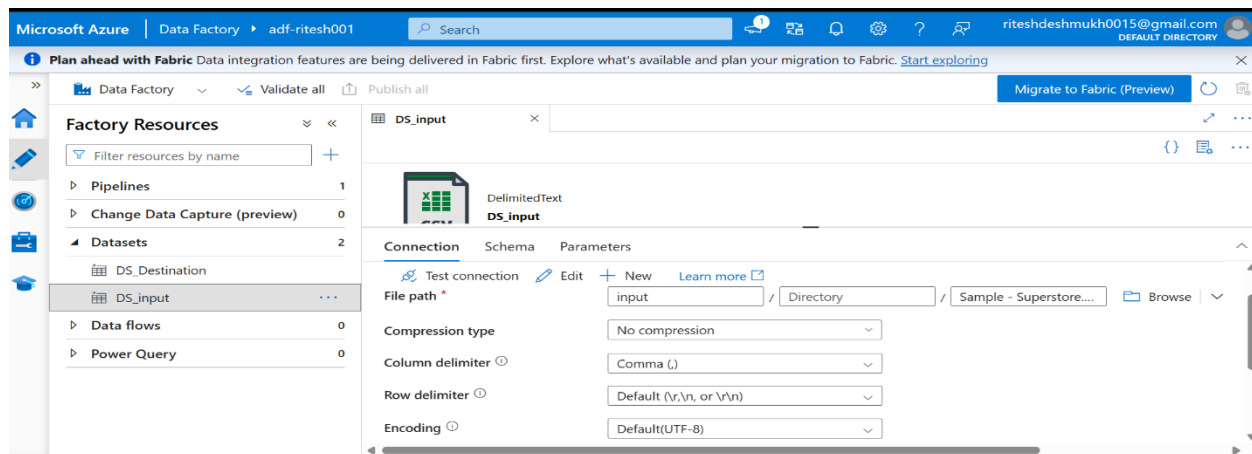
Source Dataset

The source dataset was configured using the uploaded CSV file stored in the **input** Blob container.

Configuration:

- Container: input
- File: Sample - Superstore.csv
- Format: Delimited Text

Screenshot:



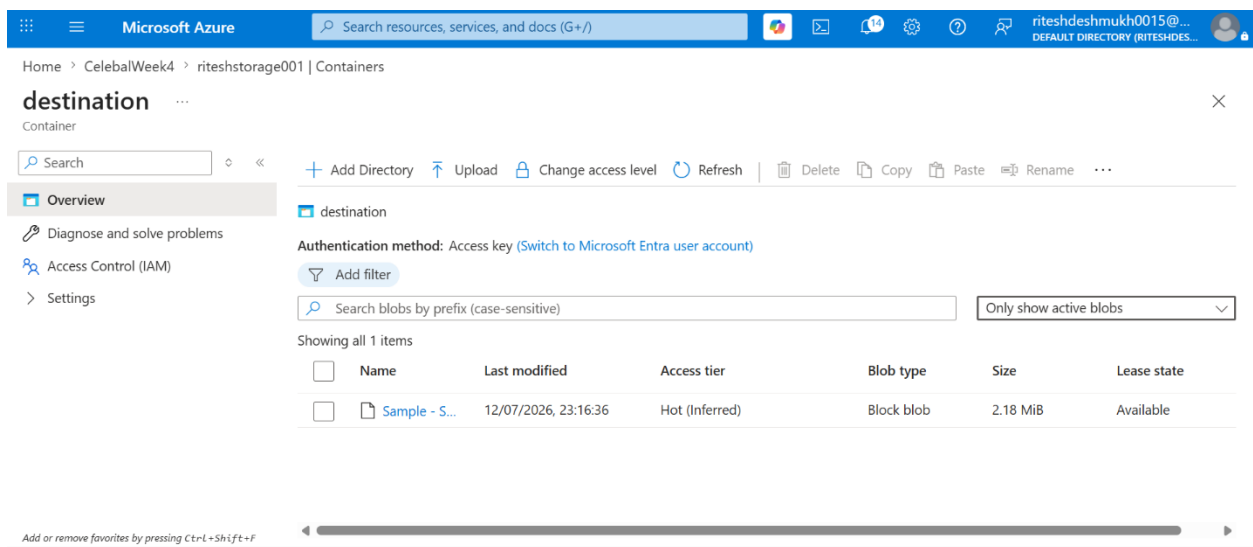
Destination Dataset

The destination dataset was configured to store the copied CSV file inside the **destination** Blob container.

Configuration:

- Container: destination
- File: Sample-Superstore_Copy.csv
- Format: Delimited Text

Screenshot:



Get Metadata Activity

The Get Metadata activity was added to the pipeline to retrieve information about the source file before copying.

Metadata Retrieved:

- Exists
- Size
- Last Modified

Task 4: Pipeline Development

A pipeline named **Pipeline_CopyData** was created.

The pipeline contains the following activities:

- Get Metadata
- Copy Data

The Copy Data activity copies the CSV file from the **input** container to the **destination** container.

Pipeline Workflow

Sample - Superstore.csv

|



Input Blob Container

|



Linked Service

|



Source Dataset

|



Get Metadata

|



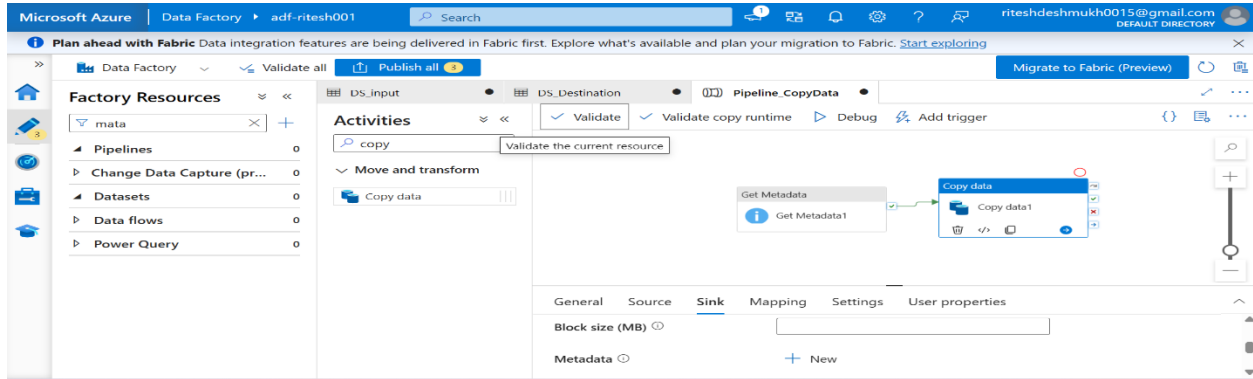
Copy Data

|



Destination Blob Container

Sample-Superstore_Copy.csv



Task 5: Pipeline Execution

The pipeline was validated successfully.

Validation Message:

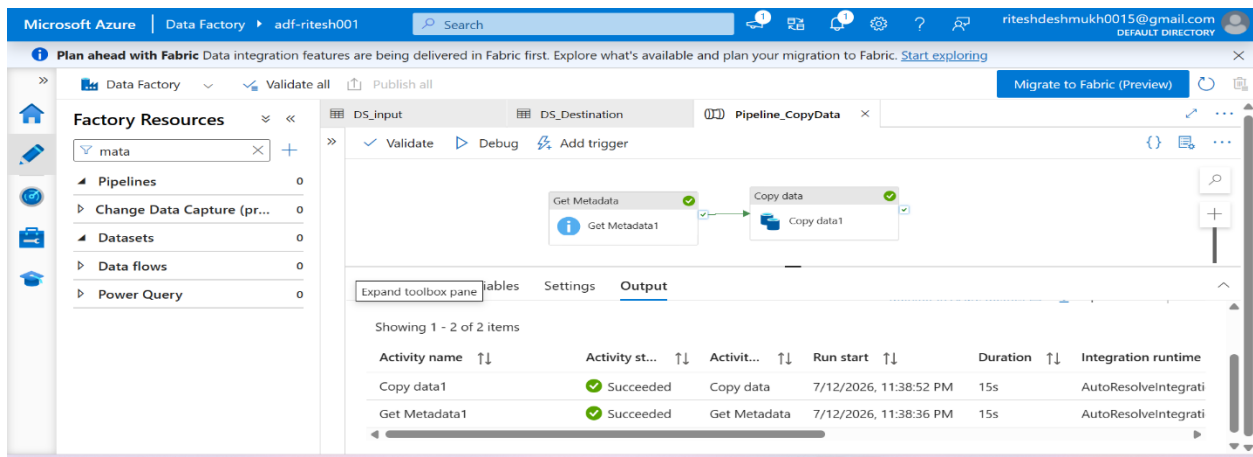
Your pipeline has been validated. No errors were found.

The pipeline was executed using the **Debug** option.

Execution Status:

- Get Metadata – Succeeded
- Copy Data – Succeeded

Screenshot:



Task 6: IAM Role Assignment

To enable secure communication between Azure Data Factory and Azure Storage, the following roles were assigned:

- Reader
- Contributor
- Storage Blob Data Contributor

Screenshot:

The first screenshot shows the 'Access Control (IAM)' page for the storage account 'riteshstorage001'. The left sidebar contains navigation links: Overview, Activity log, Tags, Diagnose and solve problems, Access Control (IAM) (selected), Data migration, Events, Storage browser, Storage Mover, and Partner solutions. The main content area has tabs for 'Check access', 'Role assignments', 'Roles', and 'Deny assignments'. Under 'Check access', there are sections for 'My access' (with a 'View my access' button) and 'Check access' (with a 'Check access' button). At the bottom, there are two cards: 'Grant access to this resource' and 'View access to this resource'.

The second screenshot shows the 'Add role assignment' page. The 'Role' tab is selected, and the 'Members' sub-tab is active. The 'Selected role' is 'Storage Blob Data Contributor'. Under 'Assign access to', the 'Managed identity' option is selected. The 'Members' section shows a table with columns 'Name' and 'Object ID', and a message 'No members selected'. At the bottom, there are buttons for 'Review + assign', 'Previous', and 'Next'. A modal window titled 'Select managed identities' is open on the right. It displays a warning: 'Some results might be hidden due to your ABAC condition.' Below this, there are dropdowns for 'Subscription' (set to 'Azure for Students') and 'Managed identity' (set to 'Data factory (V2) (1)'). A search bar is present. Below the search bar, a list of managed identities is shown, including 'adf-ritesh001' with its full object ID. At the bottom of the modal, there are buttons for 'Select' and 'Close', and a 'Feedback' link.

Output Verification

After successful execution, the copied CSV file was available in the **destination** Blob container.

Output File:

Sample-Superstore_Copy.csv

Metadata validation was completed successfully before the file was copied.

The screenshot shows the Microsoft Azure Data Factory console. The left sidebar displays 'Factory Resources' with categories like Pipelines, Datasets, and Data flows. The main area shows a pipeline named 'Pipeline_CopyData' with two activities: 'Get Metadata1' and 'Copy data1'. Both activities are marked as 'Succeeded'. Below the pipeline diagram, the 'Output' tab is selected, showing a table of activity results.

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Copy data1	✓ Succeeded	Copy data	7/12/2026, 11:38:52 PM	15s	AutoResolveIntegrati
Get Metadata1	✓ Succeeded	Get Metadata	7/12/2026, 11:38:36 PM	15s	AutoResolveIntegrati

The screenshot shows the Microsoft Azure Blob Storage console. The left sidebar displays 'Overview' with options like 'Diagnose and solve problems', 'Access Control (IAM)', and 'Settings'. The main area shows the 'destination' container. Below the container name, there is a search bar and a table of blobs.

Name	Last modified	Access tier	Blob type	Size	Lease state
Sample - S...	12/07/2026, 23:16:36	Hot (Inferred)	Block blob	2.18 MiB	Available

Result

The Azure Data Factory pipeline executed successfully.

The Get Metadata activity validated the source file, and the Copy Data activity copied the dataset from the **input** Blob container to the **destination** Blob container without any errors.

Learning Outcomes

During this assignment, the following concepts were learned:

- Azure Resource Management
 - Azure Storage Account
 - Azure Blob Storage
 - Azure Data Factory
 - Linked Services
 - Source and Destination Datasets
 - Get Metadata Activity
 - Copy Data Activity
 - Pipeline Validation
 - Pipeline Monitoring
 - Azure IAM (RBAC)
 - End-to-End Data Pipeline Development
-

Conclusion

This assignment successfully demonstrated the implementation of an end-to-end Azure data pipeline using Azure Blob Storage and Azure Data Factory. The dataset stored in the **input** Blob container was validated using the Get Metadata activity and copied to the **destination** Blob container using the Copy Data activity. The pipeline executed successfully, and the results were verified through Azure Data Factory monitoring and Blob Storage. This assignment provided practical knowledge of Azure cloud services and cloud-based data integration.